

```
1 import math
2 def entropy(data):
3     labels = [row[-1] for row in data]
4     total = len(labels)
5     counts = {}
6     for label in labels:
7         counts[label] = counts.get(label, 0) + 1
8     return -sum((count / total) * math.log2(count / total) for count in
9 counts.values())
10 def split_data(data, attribute_index, value):
11     return [row[:attribute_index] + row[attribute_index+1:] for row in data if
12 row[attribute_index] == value]
13 def info_gain(data, attribute_index):
14     total_entropy = entropy(data)
15     values = set(row[attribute_index] for row in data)
16     total = len(data)
17
18     sub_entropy = 0.0
19     for val in values:
20         subset = [row for row in data if row[attribute_index] == val]
21         sub_entropy += (len(subset) / total) * entropy(subset)
22
23     return total_entropy - sub_entropy
24 def id3(data, features):
```

```
24 def id3(data, features):
25     labels = [row[-1] for row in data]
26     if len(set(labels)) == 1:
27         return labels[0]
28     if len(features) == 0:
29         return max(set(labels), key=labels.count)
30
31     best_feat_idx = max(range(len(features)), key=lambda i: info_gain(data, i))
32     best_feature = features[best_feat_idx]
33
34     tree = {best_feature: {}}
35     remaining_features = [f for i, f in enumerate(features) if i != best_feat_idx]
36
37     feature_values = set(row[best_feat_idx] for row in data)
38     for val in feature_values:
39         subset = split_data(data, best_feat_idx, val)
40         tree[best_feature][val] = id3(subset, remaining_features)
41
42     return tree
43 # Example Usage
44 if __name__ == "__main__":
45     dataset = [
46         ['Sunny', 'Hot', 'High', 'Weak', 'No'],
47         ['Sunny', 'Hot', 'High', 'Strong', 'No'],
48         ['Overcast', 'Hot', 'High', 'Weak', 'Yes'],
```

```
45 dataset = [  
46     ['Sunny', 'Hot', 'High', 'Weak', 'No'],  
47     ['Sunny', 'Hot', 'High', 'Strong', 'No'],  
48     ['Overcast', 'Hot', 'High', 'Weak', 'Yes'],  
49     ['Rain', 'Mild', 'High', 'Weak', 'Yes'],  
50     ['Rain', 'Cool', 'Normal', 'Weak', 'Yes']  
51 ]  
52 features = ['Outlook', 'Temperature', 'Humidity', 'Wind']  
53  
54 tree = id3(dataset, features)  
55 print("Generated ID3 Decision Tree:\n", tree)  
56 |
```

Program 7: Decision Tree Learning using ID3

Generated ID3 Decision Tree:

```
{ 'Outlook': { 'Overcast': 'Yes',  
              'Sunny': { 'Humidity': { 'High': 'No', 'Normal': 'Yes'},  
                        'Wind': { 'Weak': 'No', 'Strong': 'No'}},  
              'Rain': 'Yes'}}
```

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